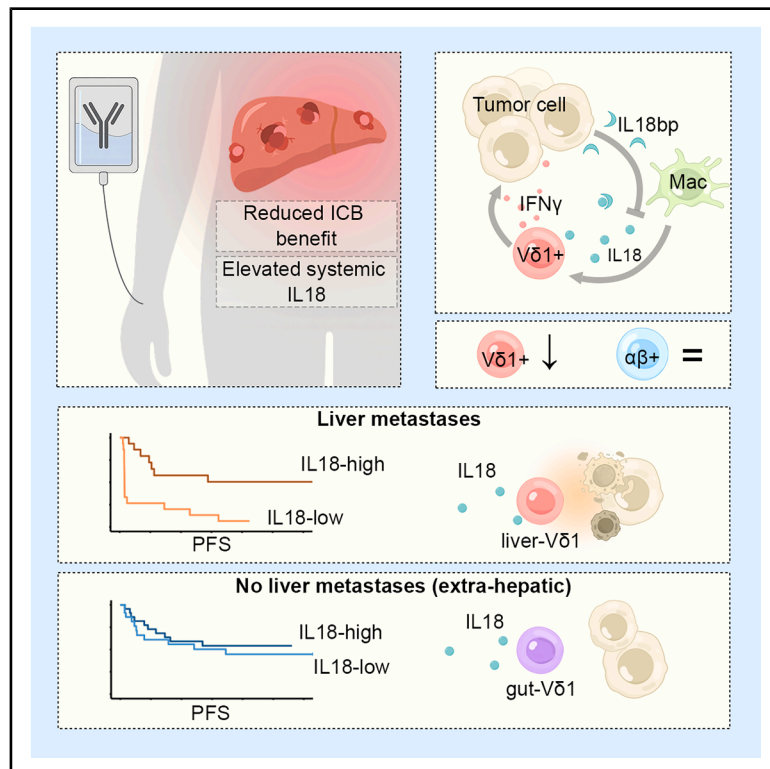


Liver metastases dampen IL18-driven $\gamma\delta$ T cell activity and immunotherapy responsiveness in colorectal cancer

Graphical abstract



Authors

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In brief

van Renterghem et al. investigate the impact of liver metastases on immunity in MMR-d/MSI-hi patients treated with ICB. They show that IL18 activates liver-derived Vδ1⁺ T cells and predicts ICB benefit selectively in LM patients. Tumor cells counteracted this response by secreting IL18BP; blockade of IL1BP restores Vδ1⁺-mediated tumor killing.

Highlights

- Patients with LMs show reduced ICB responses, with fewer infiltrating Vδ1⁺ T cells
- IL18 drives anti-tumor activity of liver Vδ1⁺ T cells, not of gut-derived counterparts
- IL18 expression predicts ICB benefit in LM patients, but not in non-LM patients
- Cancer cells secrete IL18BP to suppress IL-18 immunity; blockade restores Vδ1⁺ function



Article

Liver metastases dampen IL18-driven $\gamma\delta$ T cell activity and immunotherapy responsiveness in colorectal cancer

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SUMMARY

Patients with liver metastases (LMs) derive less benefit from immune checkpoint blockade (ICB), yet the mechanism remains poorly understood. In the liver tumor microenvironment of patients with mismatch repair-deficient (MMR-d) cancers treated with immunotherapy, we observe a reduction of V δ 1⁺ $\gamma\delta$ T cells. Hepatic V δ 1⁺ T cells express high levels of IFN γ at baseline compared to other organs. In patients with LMs, we identify elevated systemic IL18 levels compared to metastatic patients without LMs and find that its intratumoral expression is associated with ICB success exclusively in patients with LMs. While liver $\gamma\delta$ T cells are specifically sensitive to IL18 stimulation *ex vivo*, cancer cells counteract IL18-driven immunity by secretion of IL18 binding protein (IL18BP). Blockade of IL18BP enhances interferon (IFN) γ -driven immunity against organoids *in vitro*. Taken together, we identify the IL18/IL18BP/V δ 1⁺ axis as an important regulator of ICB response and a therapeutic vulnerability for patients with LMs of MMR-d tumors.

INTRODUCTION

The liver is a frontline immunological barrier with a highly specialized immune microenvironment to prevent excessive inflammation against non-pathogenic food- and microbiome-derived antigens from the intestines.^{1–3} Hepatic tolerance mechanisms, such as diminished T cell priming, have been shown to impact autoimmune diseases, viral infections, and organ transplantation.^{4–6} However, the hepatic immunoregulatory mechanisms that control the balance between homeostatic tolerance and immunological control are poorly understood.^{7,8} Importantly, the liver is also one of the most frequent sites of cancer metastasis, with colorectal cancer (CRC) being one of the most common primary cancers that spreads to the liver.^{9,10}

Immune checkpoint blockade (ICB) directed against programmed death-1 (PD-1) and programmed death-ligand 1 (PD-L1) has emerged as an effective therapy for patients with a variety of tumor types. However, accumulating evidence suggests that the presence of liver metastases (LMs) may negatively impact immunotherapeutic responses in melanoma, lung cancer, and kidney cancer.¹¹ Mechanistic studies using mouse models showed that LMs modulate systemic adaptive antitumor immunity through the elimination of $\alpha\beta$ T cells¹¹ or the generation of regulatory T cells,¹² creating a systemic “immune desert” affecting all tumor lesions throughout the body. However, given that ICB is highly effective in mismatch repair-deficient (MMR-d) CRC by

engaging both innate and adaptive immune immunity, it remains unclear whether LMs have a similar influence on treatment outcomes in this setting.^{13–19}

Recently, tissue-resident gamma delta T cells ($\gamma\delta$ T cells) have emerged as unexpected effectors of immunotherapy, acting as a cellular backup system in settings where there is reduced $\alpha\beta$ T cell immunosurveillance, such as MMR-d tumors with human leukocyte antigen (HLA) defects and melanomas with low neoantigen loads.^{13,20} Different subsets of $\gamma\delta$ T cells are defined by their T cell receptor (TCR) δ chain, of which those expressing V δ 1 and V δ 3 are primarily “tissue resident” and those expressing V δ 2 are mainly found in blood.²¹ In the context of cancer, specifically V δ 1⁺ and V δ 3⁺ T cell subsets can infiltrate tumors, have cytotoxic effects, and can be directly reinvigorated by anti-PD1 immunotherapy.^{13,15,20,22–24} However, the inflammatory signals regulating their activation and functionality in tissue—such as stimulation of the $\gamma\delta$ TCR or innate or cytokine receptors—are incompletely understood. Given the fact the liver is home to a diverse pool of tissue-resident innate lymphocytes, and the reported dampening of $\alpha\beta$ T cell immunity in the liver, we set out to delineate the contribution of $\gamma\delta$ T cells to anti-tumor immunity in the liver tumor microenvironment (TME) and the inflammatory signals governing their activity.

We utilized a cohort of patients with metastatic MMR-d cancers treated with ICB ($n = 167$ patients). This is a tumor type with a highly active immune response driven by innate lymphocyte subsets



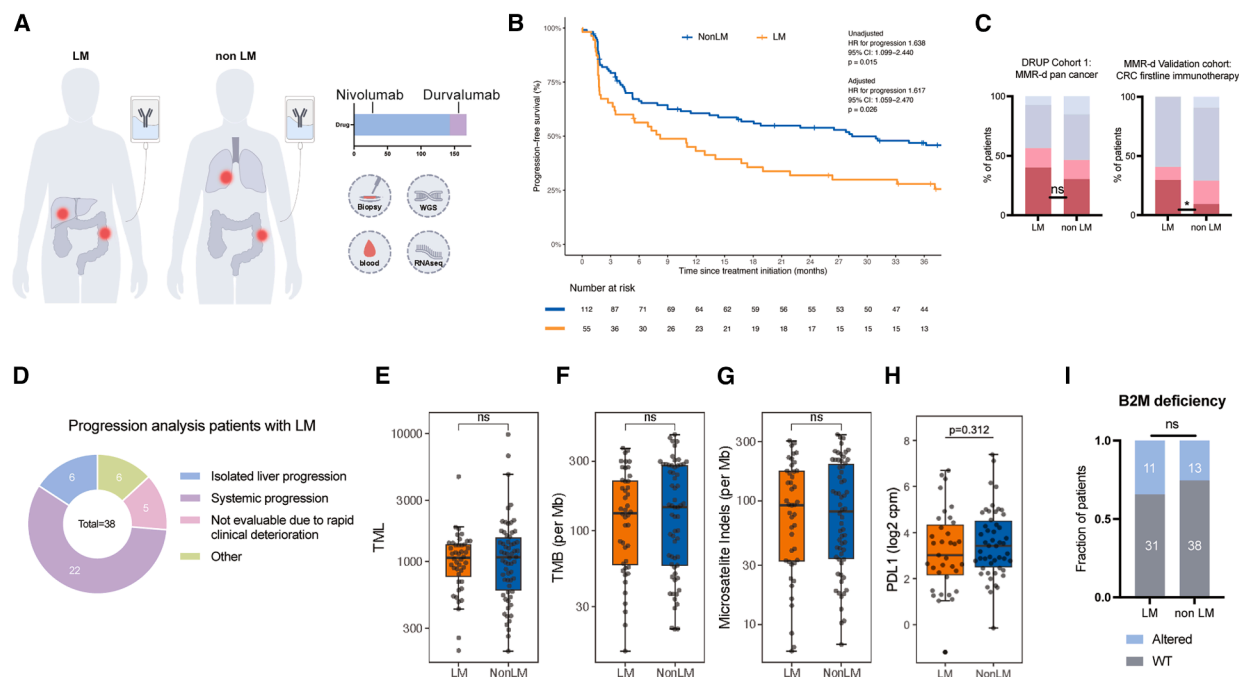


Figure 1. Patients with LMs of MMR-d cancers derive less clinical benefit of immunotherapy

(A) Cohort description of patients with advanced MMR-d tumors who received nivolumab or durvalumab and sampling strategy. LM patients (left) have metastatic cancer with liver-metastases present at baseline. Non-LM (right) are patients with metastatic cancer, with extrahepatic metastases only. The insets denote the ICB treatment; dark blue, nivolumab (anti-PD1, nivo); purple, durvalumab (anti-PD-L1, durva).

(B) Progression-free survival stratified according to the presence or absence of LMs at the start of therapy (LM, orange; non-LM, blue).

(C) RECIST1.1-defined objective responses of patients with LMs (left) and without LMs (right). Physician's assessed response to treatment in first-line real-world immunotherapy CRC cohort. CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease. p values were calculated using a two-tailed Fischer's exact test.

(D) Radiological progression pattern of patients with LMs who ended treatment due to disease progression. Pattern of progression is denoted.

(E) Tumor mutational load and LM status. Wilcoxon rank-sum test-based two-sided p value is shown. Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively.

(F) Same as (E), but showing tumor mutational burden.

(G) Same as (E), but showing microsatellite indels.

(H) PD-L1 expression based on RNA sequencing of tumor biopsies (LM, orange; non-LM, blue). Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively. Two-sided limma-voom-based regression.

(I) Presence of inactivating B2M mutations in LM and non-LM populations. The p value was calculated using a two-sided Fisher's exact test.

such as $\gamma\delta$ T cells and natural killer (NK) cells.^{14,15,25} In brief, we show that patients with MMR-d LMs display worse outcome to immunotherapy and uncover a critical role for the IL18/IL18BP-induced immunity of resident $\gamma\delta$ T cells. In comparison, we found that mismatch repair-proficient (MMR-p) CRCs ($n = 388$), which are non-responsive to ICB and have a high degree of liver tropism, present with lower intratumoral IL18 activity.

RESULTS

Liver metastases are associated with worse outcome to immunotherapy in patients with advanced MMR-d cancers

To confirm the impact of MMR-d LMs on responses to ICB, we evaluated responses to PD-(L)1 blockade (nivolumab/durvalumab) therapy in a tumor-agnostic cohort of 167 patients with metastatic MMR-d cancers. All patients were prospectively treated as part of the Drug Rediscovery Protocol (DRUP),²⁶ a multi-center clinical trial in which patients with metastatic cancer get treated

based on the molecular profiles of their tumors. Patients were subsequently divided into two groups: with or without LMs at start of ICB (Figure 1A). The presence of LMs was associated with worse progression-free survival (PFS) (hazard ratio [HR], 1.638; confidence interval (CI), 1.099–2.440; $p = 0.015$) compared to metastatic patients without LMs (Figures 1B and 1C). This result fits with the ICB response patterns observed in non-small-cell lung cancer (NSCLC) and melanoma, where the presence of LMs was also associated with a lack of response immunotherapy.¹¹ In a multivariate analysis, we found that the effect of LM was associated with PFS independently of tumor burden, previous treatment lines, primary tumor type, and drug used (Figure S1A; adjusted HR, 1.62; CI, 1.06–2.47; $p = 0.026$; see Table 1 for further clinical characteristics). The median number of previous systemic therapy lines did not differ between subgroups. Importantly, once Response Evaluation Criteria in Solid Tumors (RECIST) progression was confirmed, the vast majority of patients with LMs displayed progression of all lesions, rather than isolated progression of hepatic tumor lesions (Figure 1D).

Table 1. Baseline characteristics of MMR-d DRUP cohort

	LM (n = 55)	Non-LM (n = 112)	p value
Gender M/F (%)	29 (52.7)/26 (47.4)	42 (37.5)/70 (62.5)	0.069
Age (median [IQR])	67.00 [56.00, 74.50]	66.50 [56.75, 73.00]	0.657
Tumor (%)			
Ampullary cancer	1 (1.8)	0 (0.0)	0.013
Astrocytoma WHO grade 4	0 (0.0)	6 (5.4)	
Bile duct cancer	6 (10.9)	1 (0.9)	
Breast cancer	1 (1.8)	4 (3.6)	
Carcinoma of unknown primary	0 (0.0)	2 (1.8)	
Cervical cancer	0 (0.0)	1 (0.9)	
Colorectal cancer	30 (54.5)	53 (47.3)	
Endometrial cancer	2 (3.6)	17 (15.2)	
Gastroesophageal cancer	4 (7.3)	10 (8.9)	
Islet cell tumor	0 (0.0)	1 (0.9)	
Neuroendocrine carcinoma	0 (0.0)	1 (0.9)	
Pancreatic cancer	1 (1.8)	1 (0.9)	
Parathyroid cancer	0 (0.0)	1 (0.9)	
Prostate cancer	6 (10.9)	5 (4.5)	
Sarcoma: rhabdomyosarcoma	1 (1.8)	0 (0.0)	
Small intestine cancer	3 (5.5)	6 (5.4)	
Thyroid cancer	0 (0.0)	1 (0.9)	
Bladder and urethral cancer	0 (0.0)	2 (1.8)	
Drug = durvalumab/nivolumab (%)	11/44 (20.0/80.0)	13/99 (11.6/88.4)	0.163
WHO (%) (0.6% missing)			
0	17 (30.9)	33 (29.7)	0.832
1	35 (63.6)	74 (66.7)	
2	3 (5.5)	4 (3.6)	
Previous systemic lines (median [IQR])	1.00 [1.00, 1.50]	1.00 [0.00, 1.25]	0.213
Baseline TL tumor size (median [IQR])	91.00 [60.75, 146.00]	58.00 [38.50, 94.50]	<0.001

F, female; IQR, interquartile range; M, male; TL, target lesion; WHO, World Health Organization.

Baseline characteristics of the included patients, stratified by liver metastasis/non-liver metastasis status. Categorical variables are compared using Fisher's exact test, and continuous variables are compared using the Wilcoxon test. The WHO performance status is compared using the linear-by-linear association test.

To confirm these observations in an independent cohort of MMR-d cancers, and to highlight that the effect is generalizable independently of primary tumor type or previous treatment lines, we retrospectively assessed a "real-world" cohort of 154 patients with metastatic MMR-d CRCs from the Dutch Cancer Registry, who were treated with immunotherapy as first-line standard-of-care treatment, outside of study context. In this real-world dataset, we similarly found that patients with LMs at baseline showed less clinical benefit of ICB (clinical benefit rate [CBR] [LM/non-LM], 70.4%/90.7%; odds ratio [OR], 0.26; $p = 0.0026$) with 29.6% (LM) and 9.3% (non-LM) having progressive disease as best response to therapy (Figure 1C, right). For the main exploratory cohort, a similar trend with reduced CBR rates was visible, although this did not reach statistical significance (CBR [LM/non-LM], 60%/69.6%; OR, 0.66; $p = 0.22$). Moreover, analysis of a small cohort of patients with MMR-p metastatic CRC (mCRC, $n = 28$) treated with immunotherapy in the DRUP trial²⁶ revealed dramatically lower response rates,

with a CBR of 13.6% and no confirmed objective responses among evaluable patients. Notably, MMR-p cancers had a much higher rate of LMs (78%) compared to the MMR-d population (DRUP cohort, 32.5%; first-line CRC cohort, 38.7%). Due to the small sample size, with only 4 evaluable patients without LMs, subgroup analyses for differences in ICB efficacy were not possible. Together, this suggests that MMR-p cancers may elicit diminished hepatic anti-metastatic immunity compared to MMR-d cancers, highlighting the relevance of understanding the immunological differences between the two subtypes.

As genomic aberrations may shape metastatic capacity and tropism, as well as immunotherapy responses, we hypothesized that tumor cell-intrinsic characteristics may differ between the LM and non-LM groups of MMR-d patients. Therefore, we first assessed tumor cell-intrinsic markers of immunogenicity via whole-genome sequencing (WGS) between patients with and without LMs at the start of therapy. Notably, there was no difference in conventional genomic biomarkers of response such as

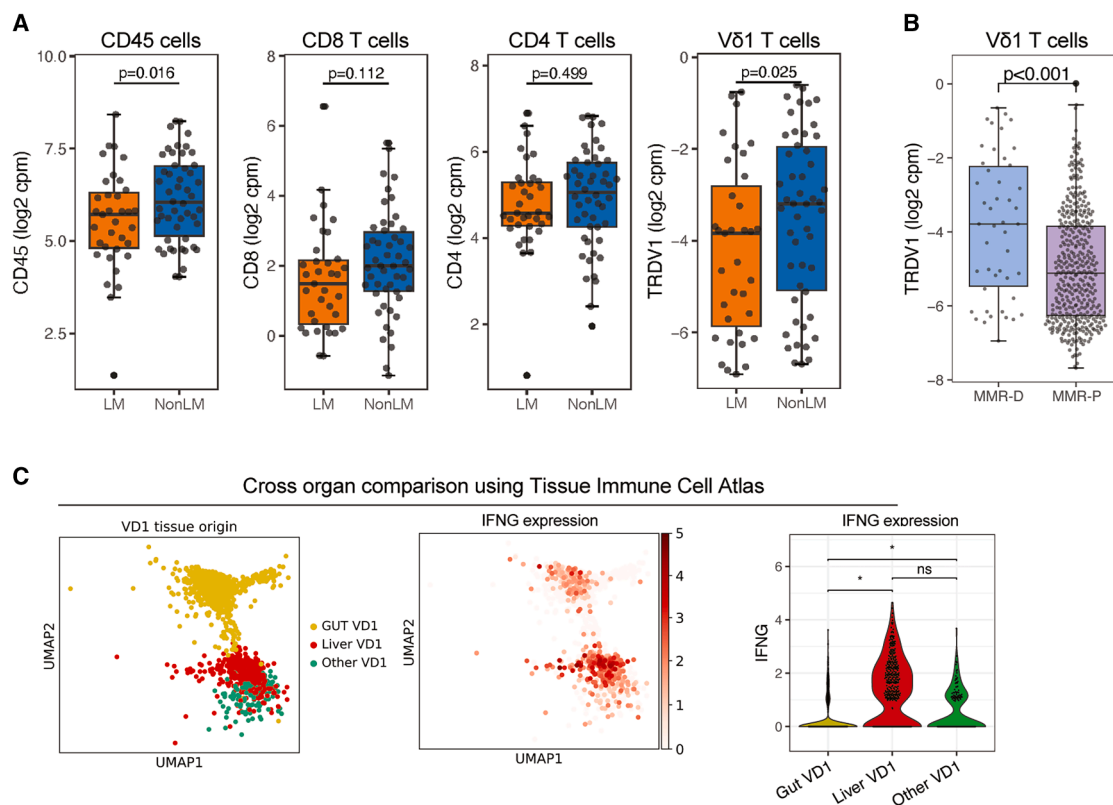


Figure 2. Vδ1⁺ γδ T cells are depleted in the liver-TME and display IFNγ-producing activity

(A) Immune marker gene set expression in LM and non-LM tumors, for CD45 cells, CD8 T cells, CD4 T cells, and Vδ1⁺ T cells. Two-sided limma-voom-based regression (LM, orange; non-LM, blue).

(B) Comparing expression of TRDV1 between MMR-d ($n = 46$) and MMR-p ($n = 388$) mCRC from the Hartwig Medical Foundation database. Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively. Two-sided limma-voom-based regression.

(C) (Left) UMAP depicting Vδ1⁺ T cells using tissue immune cell atlas database; color denotes organ of origin (gut cells, yellow; liver cells, red; other (bone marrow and lung), green). (Middle and right) IFNγ expression is shown on UMAP and according to Vδ1 localization. y axis denotes normalized counts. Dots represent individual datapoints.

tumor mutational burden, tumor mutational load, and number of microsatellite indels (Figures 1E–1G). Similarly, we did not identify differences in PD-L1 expression (based on RNA sequencing) or presence of inactivating mutations in β2-microglobulin (B2M) between the two subgroups (Figures 1H and 1I). Thus, the observed difference in immunotherapy responsiveness is likely independent of tumor cell-intrinsic antigenicity.

Vδ1 TCR expression is decreased in the liver tumor microenvironment

As we could not identify genomic differences between LM and non-LM MMR-d patients as a potential explanation for the difference in treatment response, we next investigated the cellular composition of the TME of these patient subgroups. To this end, we used bulk RNA expression data from tumor biopsies generated from tumor biopsies from patients in our MMR-d immunotherapy cohort, to estimate differences in immune cell infiltration between LM and non-LM MMR-d cancers. For the LM group, biopsies were taken from patients who displayed LMs at the start of study. The majority of biopsies were taken from the liver lesion, allowing us to estimate local immune infiltra-

tion in the liver TME (Figure S2A). As a comparison group, tumor biopsies from alternative metastatic sites from patients without LMs at baseline were used (“non-LM”). Recently, it was shown that hepatic macrophages eliminate conventional T cells in ICB-treated patients.¹¹ We, therefore, first assessed T cell infiltration of LM and non-LM cancers in our patient cohort. However, no significant differences were observed for the presence of CD4 and CD8 T cells (Figure 2A) between patients with and without LMs. Similarly, no differences in absolute lymphocyte counts of peripheral blood were identified between the two subgroups (Figure S2B). As we have recently shown that γδ T cell-mediated innate immunity is central to ICB responses in MMR-d tumors,^{13,14} we, therefore, assessed the expression of γδ T cell markers in LM and non-LM cancers. Notably, we identified a decrease of Vδ1⁺ γδ T cells (reflected by a lower expression of the TRDV1 gene) in patients with LMs compared to patients with no LM (Figure 2A). This gene is a widely accepted surrogate for Vδ1⁺ cells.^{13,22,23,27} Notably, TRDV2 and TRDV3 genes, encoding the Vδ2 and Vδ3 TCR subunits, were not found to be different between the two subgroups. To compare the TME of metastatic MMR-d and MMR-p cancers, we analyzed patients

with MMR-p ($n = 388$) and MMR-d ($n = 46$) mCRC in the Hartwig Medical Foundation database²⁸ and observed lower expression of TRDV1 in the non-responsive MMR-p population (Figure 2B). Hypothesizing that $\gamma\delta$ T cells may play a role in anti-LM immunity, we sought to further characterize and understand their function in the liver TME.

As $V\delta 1^+$ $\gamma\delta$ T cells are tissue resident, they may be adapted to their organ of origin, potentially impacting their functionality across tumor sites. Therefore, we analyzed the transcriptional phenotypes of $V\delta 1^+$ T cells across different organs by using a publicly available database, the tissue immune cell atlas.²⁹ Notably, we observed two distinct clusters based on organ-of-origin; cells derived from colon and small intestine clustered tightly together, while cells from the lung, liver, and blood clustered separately (Figure 2C). This suggests a strong divergence of the $V\delta 1^+$ T cell transcriptomes, with differential expression of cytotoxic modulators and immune checkpoints (Figure S2C). Notably, $V\delta 1^+$ T cells from the liver expressed high levels of IFN γ at baseline compared to gut-counterparts (Figure 2C, middle and right). Thus, these results show that $\gamma\delta$ T cells are transcriptionally heterogeneous across organs, suggesting that the mechanisms of activation may also differ depending on their organ of origin. In summary, we observed a depletion of $V\delta 1^+$ $\gamma\delta$ T cells in patients with LMs that is independent of B2M status (Figure 1I). Given the high expression of IFN γ of these cells at baseline, this alludes to a potential functional significance of liver- $\gamma\delta$ T cells as mediators of anti-metastatic immunity.

IL18 is associated with immunotherapy response only in patients with LMs

Given the observed reduction of $V\delta 1^+$ T cells in LMs, we set out to further investigate the inflammatory signals governing their activity. Therefore, we measured the expression of a panel of inflammatory cytokines in the plasma of patients with gastrointestinal MMR-d tumors. When comparing plasma cytokine levels between patients with and without LMs, the only significantly different and increased cytokine was IL18, which is a major regulator of Th1 and NK cell activities (Figure 3A, left, and S3A). To confirm these observations, we next measured IL18 levels in a pan-cancer MMR-d cohort and similarly observed increased IL18 levels in the LM group (Figure 3A, right). Moreover, when assessing the intratumoral expression of a panel of inflammatory cytokines (see Table S1 for gene list), IL18 was the cytokine most associated with response in the LM population (Figure 3B). Interestingly, IL18 expression was only associated with response to immunotherapy in patients with LMs, but not in the non-LM subgroup (Figure 3C), suggesting a liver-specific effect of IL18. Next, we stratified LM and non-LM subgroups based on IL18 expression and assessed their PFS. Strikingly, IL18-low LMs had significantly lower PFS compared to IL18-high LMs (Figure 3D; HR, 4.1; CI, 1.28–13.1; $p = 0.0174$), whereas patients with IL18-high LM had comparable PFS compared to patients without LMs. Moreover, IL18-high LMs displayed an enrichment of interferon response signatures compared to IL18-low LMs (Figure 3E). In comparison, intratumoral IL18 expression was significantly lower in the ICB-unresponsive MMR-p CRC population compared to MMR-d cancers (Figure 3F). Thus, the presence of LMs is associated with

changes in systemic immunity and inflammatory cytokine profiles. Additionally, the association of IL18 expression with immunotherapy response, exclusively in patients with LMs, suggests that effective immunity against LMs is particularly sensitive to reductions in intratumoral IL18 levels.

While tumor cells are capable of endogenously expressing IL18 (Figure S3B), we found no correlation between tumor burden (as a proxy for the amount of tumor cells) and serum levels of IL18 ($R = -0.007971$, $p = 0.9750$) or IL18 expression ($R = -0.158$; $p = 0.185$) (Figures S3C and S3D), suggesting a cancer cell-extrinsic source. We, therefore, assessed the expression of IL18 and NLRP3 in the liver cell atlas,³⁰ a large single-cell RNA sequencing database of human healthy liver cell populations (Figures 3G–3I). NLRP3 is part of the inflammasome protein complex that regulates IL18 levels and converts pro-IL18 to its mature form.³¹ Notably, we observed co-expression of IL18 and NLRP3 in hepatic myeloid cells, with the largest population being hepatic macrophages and monocyte-derived cells (Figure 3I). In our transcriptomic MMR-d LM patient dataset, we assessed which immune cell signatures were associated with intratumoral IL18 expression. Here, we observed a similar pattern, in which myeloid cell populations such as neutrophils, dendritic cells, and macrophages were among the gene sets showing the strongest association to IL18 (Figure 3J). Together, this suggests that in a tumor-bearing liver, hepatic myeloid cells could serve as an important source of IL18.

IL18 potentiates anti-tumor immunity of liver-derived $V\delta 1^+$ $\gamma\delta$ T cells and is dependent on IL12

Given the observed heterogeneity of $\gamma\delta$ T cells across organs, as well as the LM-specific association of IL18 with immunotherapy responsiveness, we wanted to determine whether there was an organ-specific sensitivity of $\gamma\delta$ T cells to IL18. To this end, we isolated tissue-resident immune cells (TRICs) from primary normal human colons and livers and stimulated the TRIC pool with cytokines *ex vivo* for 24 h (Figure 4A). To ensure that $\gamma\delta$ T cells retained their native tissue phenotypes, cells were not expanded after isolation but stimulated directly. Upon stimulation with IL12 and IL18, both liver and gut $V\delta 1^+$ T cells produced IFN γ . However, the reactivity of liver $V\delta 1^+$ T cells was much more pronounced compared to gut $V\delta 1^+$ T cells, with 51.66% versus 12.39% of $V\delta 1^+$ T cells producing IFN γ , respectively (Figures 4B and 4C). Additionally, upon incubation with solely IL18, only liver $V\delta 1^+$ T cells displayed increased reactivity, but gut $V\delta 1^+$ T cells did not. Looking within the liver -TRIC, liver-resident $\alpha\beta$ T cells did not show IL18-induced IFN γ production (Figure 4D). To further assess the liver specificity, we stimulated fresh $V\delta 1^+$ T cells from peripheral blood with cytokines, and similarly to the gut, we observed limited production of IFN γ (Figure S4A). The differential sensitivity between organ of origin could not be explained by altered expression of IL18R1, as this was similarly expressed on $V\delta 1^+$ T cells across organs (Figures S4B–S4D). Considering the IL18 effect depends on cooperative signaling with cytokines such as IL15 and IL12, we assessed the expression of these cytokines, which were expressed similarly between patient subgroups (Figure S4E). Thus, the liver-resident $V\delta 1^+$ T cell pool displays increased sensitivity to IL18 compared to other gut and circulating cells. These results indicate that the inflammatory programs driving peripheral $V\delta 1^+$ activity may differ

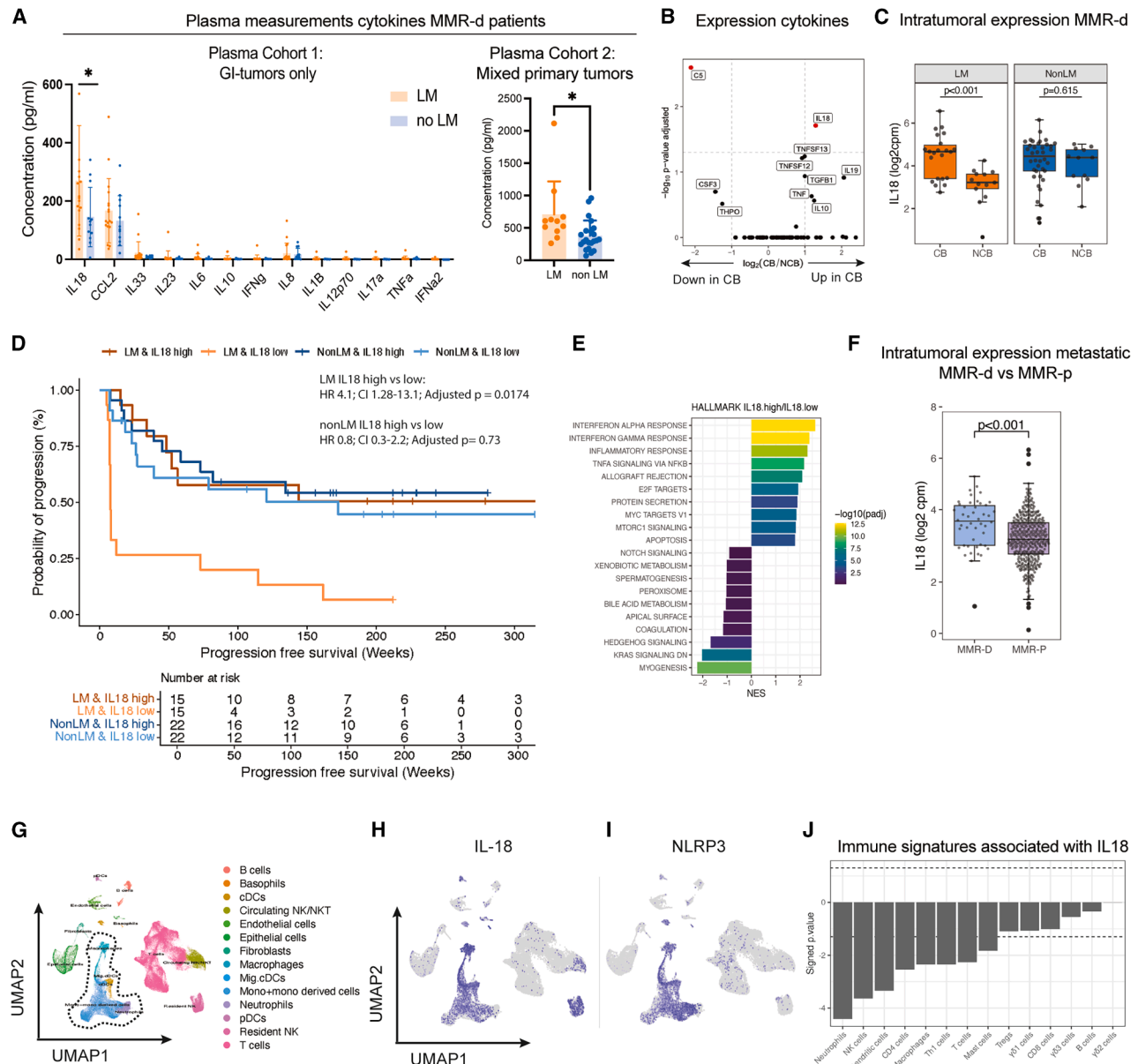


Figure 3. IL18 is specifically associated with response to immunotherapy in patients with liver metastases

(A) Cytokine levels in pre-treatment plasma from MMR-d patients stratified according to LM status (LM, orange; non-LM, blue). Cytokines were measured using a cytokine bead array; concentrations are noted in pg/mL. Individual datapoints reflect single patients. Significance was determined based on multiple unpaired t tests; asterisks indicates p value (<0.05). (Left) Gastrointestinal MMR-d tumors only, $n = 30$. (Right) Pan-cancer MMR-d cancers $n = 34$. Data are represented as mean $\pm$ SD with dots depicting individual datapoints.

(B) Volcano plot showing the differential expression of cytokine genes in patients with LMs with clinical benefit versus no clinical benefit.

(C) Expression of IL18 stratified to LM status and clinical benefit. Two-sided limma-voom-based regression. (LM, orange; non-LM, blue). CB, clinical benefit; NCB, non-clinical benefit. Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively.

(D) Progression-free survival of patient subgroups, stratified according to LMs and subclassified into IL18-high and -low expression. (LM IL18-high, dark orange; LM IL18-low, light orange; non-LM IL18-high, dark blue; non-LM IL18-low, light-blue). The p values were calculated using a multivariate cox regression model correcting for primary tumor type and drug.

(E) Gene set enrichment analyses of IL18-high versus IL18-low patients with liver metastases.

(F) Comparison of IL18 expression between MMR-d ($n = 46$) and MMR-p ($n = 388$) mCRCs from the Hartwig database. Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively.

(G–I) Liver cell atlas database was used; (G) UMAP visualization of all liver-resident cells. IL18 (H) and NLRP3 (I) expression are depicted.

(J) Association of immune cell signatures with IL18 expression in LM MMR-d patients. Bars represent the directionality (signed) $\log_{10} p$ values of association with IL18 expression. For both panels, p values are two-sided and were generated by limma-voom for individual genes and by logistic regression for gene sets.

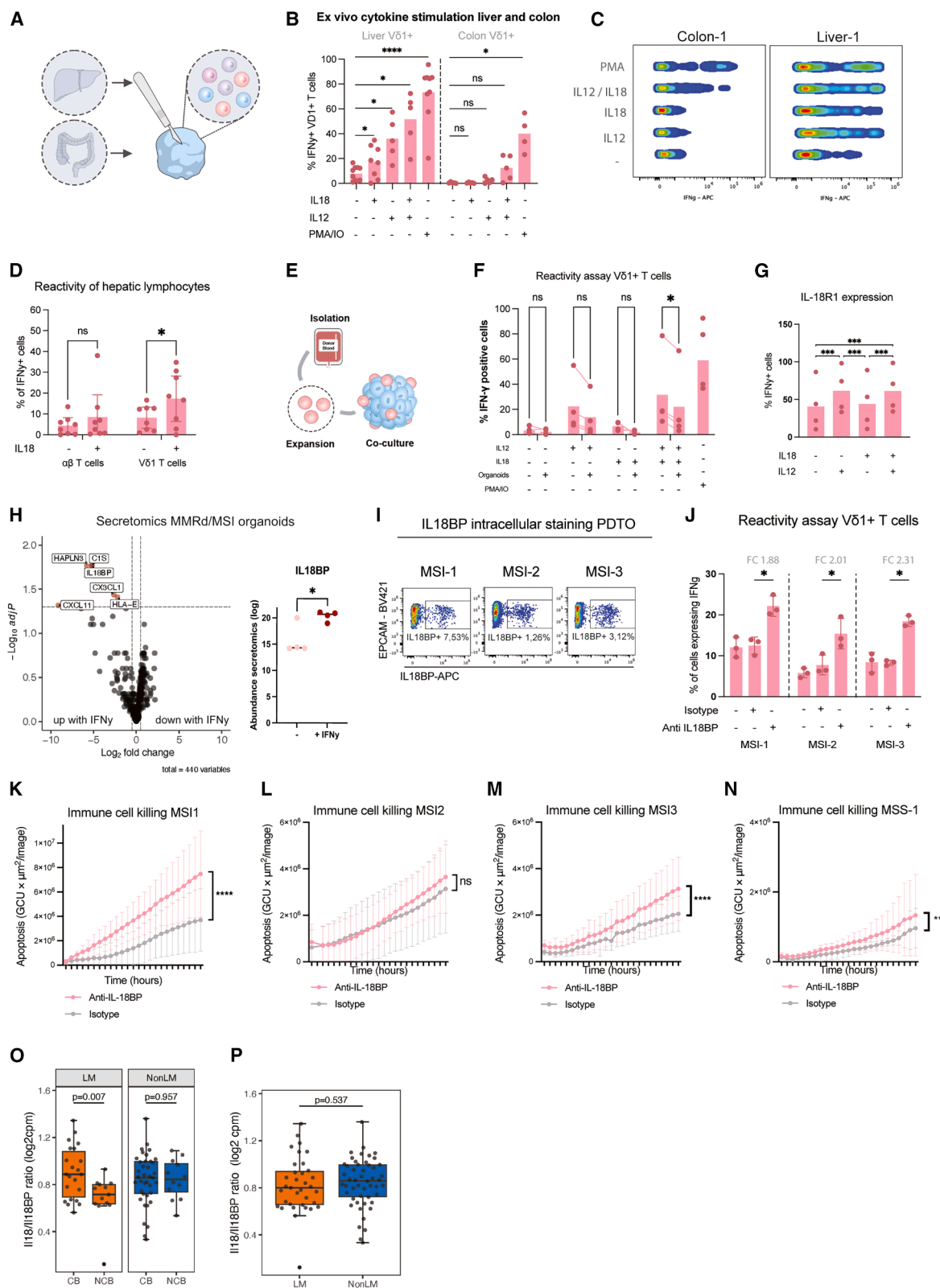


Figure 4. IL18 drives anti-tumor immunity of $\gamma\delta$ T cells and is limited by cancer cell-derived IL18BP

(A) Schematic denoting the usage of tissue-derived immune cells from primary human tissue.

(legend continued on next page)

across organs and metastatic sites, due to adaptation of these cells within their organ of origin.

To further study effects of IL18 on $\gamma\delta$ T cells and how this may subsequently impact anti-tumor immunity, we employed an *in vitro* system using peripheral blood mononuclear cell (PBMC)-derived expanded $V\delta 1^+$ $\gamma\delta$ T cells (Figure 4E). While fresh blood-derived $V\delta 1^+$ T cells were not found to be sensitive to cytokine stimulations, after long-term expansion, we did observe sensitivity as measured by IFN γ production (Figures 4F and S4E). Unlike their tissue-resident counterpart, this depended on the addition of IL12, which led to upregulation of the IL18 receptor (Figure 4G). Next, we compared $V\delta 1^+$ T cell reactivity upon cytokine stimulation in co-culture with tumor cells and noticed a significant reduction in reactivity in the presence of cytokine stimulation (Figures 4F and S5A). Additionally, we co-cultured $V\delta 1^+$ $\gamma\delta$ T cells with two patient-derived tumor organoids (PDTOs) from MMR-d cancers and assessed viability after 24 h. When adding recombinant IL18, we observed only a modest, non-significant increase in tumor cell killing at this time point (Figure S5B). Additionally, as tumor cells can endogenously produce IL18, we performed a PDTO- $V\delta 1^+$ $\gamma\delta$ T cells co-culture in the presence of IL12 and added an IL18 blocking antibody. The addition of an IL18-blocking antibody did not alter T cell reactivity (Figure S5C). These results suggest that although tumor cells are capable of producing IL18, it may not be functionally active. Together, these data suggest that cancer cells may constrain the activity of $\gamma\delta$ T cells, prompting us to further examine the mechanistic basis underlying this immunosuppressive process.

Cancer cells limit $\gamma\delta$ T cell-mediated anti-tumor immunity by secreting IL18BP

To determine potential proteins derived from tumor cells that modulate IL18 responses, we analyzed the secreted protein

fraction of four MMR-d PDTOs upon IFN γ stimulation using mass spectrometry. Notably, as one of the top upregulated proteins, we found that PDTOs strongly increase IL18BP secretion in response to IFN γ (Figure 4H). Interestingly, IL18BP is a decoy protein that intercepts IL18 and prevents signaling in cells expressing the IL18 receptor.^{32–34} IL18BP expression was confirmed via flow cytometry for a panel of three MMR-d PDTO's, where it was detected in all three lines (Figure 4I). The increased secretion of IL18BP upon IFN γ stimulation in tumor organoids was confirmed by ELISA and was not observed in a healthy colon organoid line (Figure S5D).

Next, to confirm that cancer cell-derived IL18BP can modulate immune cell reactivity, we employed an *in vitro* co-culture assay with PBMC-derived $V\delta 1^+$ $\gamma\delta$ T cells and with three PDTOs and assessed $V\delta 1^+$ $\gamma\delta$ T cell reactivity via flow cytometry based on IFN γ production. When IL18BP was blocked with an antibody, $V\delta 1^+$ $\gamma\delta$ T cells were significantly more reactive compared to isotype control in all three PDTO lines (Figure 4J). Next, we assessed whether blocking IL18BP could potentiate the killing of cancer cells. To this end, we co-cultured $V\delta 1^+$ $\gamma\delta$ T cells with three MMR-d/MSI-hi PDTO's in the presence of a fluorescent cleaved-caspase-3/7 reporter to measure cancer cell apoptosis over time. In MSI-1 and MSI-3, cancer cell apoptosis was significantly higher when anti-IL18BP was added compared to isotype control (Figures 4K–4M). To assess whether IL18BP blocking strategies could also be relevant for MMR-p/MSS tumors, we performed a killing assay with an MMR-p CRC line and observed increased killing upon IL18BP blockade (Figure 4N). Next, to assess the association of IL18BP with immunotherapy response, we calculated a ratio between IL18 and IL18BP expression for each patient. Notably, in patients with LMs, immunotherapy-responsive patients had a significantly higher IL18/IL18BP ratio

(B) Tissue-derived immune cells from liver and colon samples were isolated and stimulated with cytokines (IL18, 10 ng/mL; IL12, 10 ng/mL) for 24 h, and IFN γ production was determined via flow cytometry. Representative of 4–8 primary tissue donors. Data are represented as mean $\pm$ SD. Asterisks indicate significance (* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$; **** $p \leq 0.0001$).

(C) Representative flow cytometry plots of $V\delta 1^+$ T cell reactivity from colon and liver samples.

(D) Reactivity as measured by IFN γ production of liver-resident $V\delta 1^+$ T cells and $\alpha\beta$ T cells upon stimulation with IL18 (10 ng/mL; 24 h). Paired *t* test was used to test differences upon cytokine stimulation. Data are represented as mean $\pm$ 95% CI. Asterisks indicate significance (* $p \leq 0.05$).

(E) Schematic overview of immune-PDTO co-culture assays.

(F) Reactivity of PBMC-derived, expanded $V\delta 1^+$ $\gamma\delta$ T cells when stimulated with cytokines (IL12, 10 ng/mL; low-dose IL18, 1 ng/mL; 24 h), in the presence or absence of PDTO line (MSI-1). Data are reflective of 4 independent $V\delta 1^+$ $\gamma\delta$ T cells donors. Data are represented as mean $\pm$ SD. Asterisks indicate significance (* $p \leq 0.05$).

(G) Expression of IL18R1 on PBMC-derived $\gamma\delta$ T cells upon cytokine stimulation (IL12 10 ng/mL, 24 h). Data are reflective of 4 biological replicates. Data are represented as mean; points reflect individual measurements.

(H) (Left) Volcano plot denoting newly synthesized proteins from patient-derived tumor organoids (MMR-d; $n = 4$) in response to IFN γ stimulation (20 ng/mL for 24 h). (Right) Abundance of secreted IL18BP with or without IFN γ stimulation as in (H). Asterisks indicates *p* value (<0.05) based on paired *t* test ($n = 4$). Symbols indicate individual organoids (no IFN γ , pink; IFN γ , dark red). Paired *t* test was used to calculate differential secretion upon IFN γ stimulation.

(I) Representative flow cytometry plots of PDTO lines MSI-1, MSI-2, and MSI-3 showing intracellular IL18BP.

(J) Interference assay using PBMC-derived expanded $V\delta 1^+$ T cells ($n = 3$), co-cultured with three different MMR-d/MSI patient-derived organoids pre-stimulated with IFN γ (20 ng/mL, 24 h) and stimulated with IL12 (10 ng/mL) and low-dose IL18 (1 ng/mL). Three anti-IL18BP or isotype antibodies were added to culture to show impact on reactivity. Data are represented as mean $\pm$ SD. Asterisks indicate significance (* $p \leq 0.05$).

(K–M) Killing assay of MSI-1 (K), MSI-2 (L), and MSI-3 (M) PDTO, co-cultured with 4 different $\gamma\delta$ T cell donors. Co-culture was in the presence of a fluorescent cleaved-caspase-3/7 reporter to measure cancer cell apoptosis over time (organoid with isotype, dark gray; anti-IL18BP, red). Plot depicts mean $\pm$ SD. two-way ANOVA with Tukey multiple testing correction. Asterisks indicate significance (* $p \leq 0.05$; ** $p \leq 0.01$; *** $p \leq 0.001$; **** $p \leq 0.0001$).

(N) Killing assay of MSS-1 PDTO, cocultured with 3 different $\gamma\delta$ T cell donors. Plot depicts mean $\pm$ SD. Asterisks indicate significance (** $p \leq 0.01$).

(O) Ratio of expression of IL18 and IL18BP stratified to LM status and clinical benefit. (LM, orange; non-LM, blue). CB, clinical benefit; NCB, non-clinical benefit. Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively.

(P) Ratio of expression of IL18 and IL18BP comparing total LM and non-LM populations (LM, orange; non-LM, blue). Boxes, whiskers, and dots indicate quartiles, 1.5 interquartile ranges, and individual data points, respectively.

compared to non-responding patients (Figure 4O). For patients without LMs, IL18/IL18BP ratios were not associated with response (Figure 4O) and both the transcriptomic ratio and plasma protein levels were not significantly different from the LM group (Figures 4P and S5E). Together, these results suggest that IL18 signaling contributes to the specific activation of $V\delta 1^+$ $\gamma\delta$ T cells in the liver, which can locally be suppressed by cancer cells through the production of IL18BP. Additionally, our findings highlight that IL18-centric therapies may be beneficial for boosting $\gamma\delta$ T cell functionality in the tumor.

DISCUSSION

The liver has evolved as a central immunogenic hub of the body.^{1,2} Here, we show that the presence of LMs in metastatic MMR-d patients dampens immunotherapy response throughout the body, also affecting extrahepatic tumor sites. Importantly, our study identifies an immunological axis involving IL18/IL18BP/ $V\delta 1^+$ T cell, capable of shifting the local hepatic immune environment from suppression to control, which is significantly associated with therapeutic responses to ICB.

Previous studies reported that LMs from melanoma and NSCLC can co-opt hepatic tolerance mechanisms—specifically, macrophage-mediated elimination of $\alpha\beta$ T cells¹¹—to dampen systemic ICB responses. In contrast, within the MMR-d population, we do not observe a depletion of $\alpha\beta$ T cells intratumorally or in the peripheral blood. We have previously shown that these tumors exhibit distinct immune dynamics in response to ICB, characterized by a prominent role for innate immunity.^{13,14} This immunological divergence may be explained by the loss HLA class I expression that occurs in up to 70% of MMR-d cancers,^{35–38} leading to evasion of adaptive immunity.^{39–41} The differential immunological pressures these tumors undergo may, in turn, drive the emergence of different negative feedback mechanisms. Here, we identified a mechanism regulating the activity of liver-resident $\gamma\delta$ T cells.

$\gamma\delta$ T cells are a diverse cell type, consisting of subsets exhibiting varying functions that, in the context of cancer, range from tumor-promoting to tumoricidal effects.^{20,22,24,42} We found evidence of functional heterogeneity among $\gamma\delta$ T cells across organs and demonstrated that, in the liver, they may exhibit a specifically anti-tumorigenic, IFN γ -producing role. We showed (1) divergent transcriptomes of $V\delta 1^+$ T cells across organs, with liver- $V\delta 1$ particularly high in IFN γ expression, and (2) unequal IL18 responsiveness *ex vivo*, which is mirrored in patient ICB response data. Previously, the adult human hepatic $\gamma\delta$ T cell compartment was investigated and was found to be enriched with a tissue-resident effector subset, displaying enhanced cytokine production and sensitivity to innate stimuli compared to the blood-derived cells.⁴³ Our results align with these previous observations, as we see that $\gamma\delta$ T cells from the liver produce higher levels of IFN γ in response to cytokine stimulation compared to those derived from fresh blood and gut. Interestingly, when blood-derived $\gamma\delta$ T cells underwent expansion, sensitivity to innate stimuli such as IL12 and IL18 was increased, shifting the functional profile of these cells, similar to what is observed in the liver. These findings provide evidence that organ-specific contexts can dictate the functional state of $\gamma\delta$ T cells and impact anti-tumor effector function. These organ-

specific programs may result from local cues upon their arrival in tissues, or may already be epigenetically imprinted during tissue development, as is the case with long-lived tissue-resident macrophages, for example.^{44–46} It is currently unclear whether the organ heterogeneity in the $V\delta 1^+$ $\gamma\delta$ T cell pool is due to organ imprinting, ontogeny, or a combination of both.⁴⁷ Thus, there is a need to further understand organ-specific differences and molecular profiles of resident ICB effector populations, to rationally tailor potential therapeutic strategies to induce site-specific anti-metastatic immune responses.

More specifically, we identified IL18 and its decoy IL18BP as key regulators of immunological activity in the liver metastatic TME. Using *ex vivo* assays with TRICs, we showed that IL18 has disparate immunological effects based on anatomical site of action and found that the hepatic $\gamma\delta$ T cells are uniquely sensitive to this cytokine. While multiple cell types in the liver TME express its receptor and may thus be affected by IL18, our results pinpoint $V\delta 1^+$ $\gamma\delta$ T cells as a key target population. *Ex vivo* assays showed sensitivity of $V\delta 1^+$ $\gamma\delta$ T cells, but not $\alpha\beta$ T cells, to IL18 stimulation. Similarly, in patient tumors, we see a reduction in the expression of $V\delta 1^+$ signatures, but not $\alpha\beta$ T cells. The intratumoral expression of IL18 was strongly associated with immunotherapy response in patients with LM, suggesting that—specifically in the liver—IL18-driven immune activity is necessary for ICB-mediated immune reinvigoration. However, to formally prove the involvement of IL18 in ICB responses, one would require loss-of-function models—which is unfortunately not possible as no subset equivalent to $V\delta 1^+$ or $V\delta 3^+$ T cells has been identified in mice. Additionally, the exact source of IL18, its regulation, and spatial distribution are still important questions that need to be addressed. Secretion of IL18BP upon IFN γ release may dampen subsequent immunological activity. While IL18BP levels are likely lower in the pre-treatment microenvironment, IL18BP blockade does provide the opportunity of increasing the bioavailability of local IL18. Using PDO-immune cell co-cultures,^{48–50} we show that IL18BP blockade enhanced $\gamma\delta$ T cell reactivity and PDO killing. While this report focused on the role of IL18/IL18BP in the pre-treatment TME, it would be of interest to understand how IL18BP expression changes upon ICB treatment and shapes subsequent immune-reinvigoration dynamics. Additionally, whether the secretion of this decoy protein at the primary tumor site is involved in modulating local immunity and whether it plays a role in the formation of an immune-suppressed pre-metastatic niche in the liver is currently unclear.

Our results suggest that patients with MMR-d LMs may benefit from therapies empowering $\gamma\delta$ T cells. $\gamma\delta$ T cell therapies provide the advantage of acting independently of tumor antigen-specific recognition and, therefore, can still eliminate tumors that escaped T cell surveillance. Cytokine-based stimulation of $\gamma\delta$ T cells with IL12 and IL18 induced a potent, TCR-independent anti-tumor immune response. Therefore, modulation of the IL18 axis may be an attractive therapeutic strategy in concert with ICB or adoptive $\gamma\delta$ T cell therapies. Previously, clinical development of recombinant IL18 as a monotherapy was found to be safe and well tolerated, but it was abandoned due to lack of efficacy.⁵¹ Currently, various forms of IL18-centric immunotherapies are under development, such as blocking antibodies for IL18BP, highly potent “designer” cytokines, or cytokine-secreting cell

therapies.^{32,52,53} Our results show that such strategies could additionally help to boost the functionality of $\gamma\delta$ T cells, yet the site of action may affect the biological activity.

For the immunotherapy non-responsive MMR-p CRC population, we found lower baseline intratumoral expression of IL18. If IL18 is a central player in liver immunity, it would be of interest to see whether IL18 agonism strategies in concert with immunotherapy could induce tumor immunity for these notoriously cold tumors. Similarly, the observation that MMR-p tumors exhibit increased liver tropism compared to MMR-d tumors, even though they come from similar anatomical locations, suggests that MMR-p tumors evade triggering anti-metastatic immune responses when entering the liver. It is tempting to speculate that the balance between IL18/IL18BP could determine innate immune recognition and act as a tissue-specific barrier to disseminated tumor cell outgrowth. Evaluating the potential of decoy-resistant IL18 agonists to prevent the development of metachronous LM is, therefore, of interest.

In conclusion, our results implicate the IL18/IL18BP axis as a major barrier for ICB responses in patients with MMR-d cancers that have metastasized to the liver. Mechanistically, liver-resident $\gamma\delta$ T cells act as key tissue sentinels responding to IL18 by secretion of IFN γ . Cancer cells may protect themselves by subsequently secreting IL18BP, which acts as a negative feedback mechanism to mute IL18-induced anti-tumor immunity. Conceptually, our results demonstrate that resident immune effector populations may differ intrinsically based on organ of origin, thereby shaping anti-tumor immunity within the metastatic site. Thus, metastatic cancer is a systemic malady that requires therapeutic strategies beyond the primary disease site to effectively combat metastases.

Limitations of the study

To further reinforce the association between LMs and ICB response in MMR-d cancers, other validation cohorts with consistent efficacy metrics and validation of MMR-d status via WGS are of interest. One limitation of our study is the reliance on RNA expression data for immune cell infiltration, specifically because the $V\delta 1^+$ TCR is lowly expressed. While this is standard practice for the infiltration of $V\delta 1^+$ T cells—as no subset-specific antibodies for FFPE exist—orthogonal ways of estimating $V\delta 1^+$ infiltration are of interest. Additionally, for comparison between LM and non LM patients, biopsies from alternative metastatic localizations were used, which may introduce bias due to organ-specific differences in the immune landscape. Additionally, *ex vivo* assays using liver-derived $\gamma\delta$ T cells were not validated with intratumoral $V\delta 1^+$ T cells, but were derived from normal liver specimens, as resection material of MMR-d LM is not readily available in the clinic. For IL18BP, the functional impact was only tested on PBMC-derived $V\delta 1^+$ T cells in co-culture with PDO's, but not liver-derived cells. The majority of PDO's were derived from primary resection material. Moreover, further understanding of the regulation of IL18 expression in the TME, including the cellular source and spatial distribution, is of interest.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to Emile Voest (e.voest@nki.nl).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- De-identified patient WGS, clinical metadata, and transcriptomics data from the Hartwig Medical Foundation are freely available for academic use through standardized procedures. Request forms can be found at <https://www.hartwigmedicalfoundation.nl/en/data/data-access-request/>. Data are publicly available as of the date of publication.
- The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE⁵⁴ partner repository with the dataset identifier PRIDE: PXD071538.
- This paper does not report original code. The bioinformatics for the WGS and transcriptomics data was performed with open-source tools. DOIs are listed in the [key resources table](#).
- Any additional information required to reanalyze data reported in this work is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

A.W.J.v.R., conceptualization, formal analysis, investigation, visualization, methodology, writing – original draft, and writing – review and editing; M.P.-M., data curation, formal analysis, investigation, visualization, methodology, and editing; M.W., data curation, formal analysis, and investigation; K.V., data curation and formal analysis; M.S., investigation; L.J.Z., data curation; K.K.D., conceptualization and supervision. L.A., conceptualization, supervision, and editing original draft. E.E.V., conceptualization, supervision, funding acquisition, and writing – original draft.

DECLARATION OF INTERESTS

E.E.V. is founder and current member of the supervisory board of the Hartwig Medical Foundation, independent non-executive director of Sanofi, co-founder of Mosaic Therapeutics, and board member and founder of the Center for Personalized Cancer Treatment. He has received clinical study grants from Amgen, AstraZeneca, BI, BMS, Clovis, Eli Lilly, GSK, Ipsen, MSD, Novartis, Pfizer, Roche, and Sanofi. K.K.D. provided consultancy services to Achilles Therapeutics UK Ltd.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT5.0 in order to edit and review text, as well as edit their own illustrations for the graphical abstract. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
BUV805, Mouse Anti-human CD4	BD biosciences	Cat#612888/612887; RRID:AB_2870176
BUV563, Mouse Anti-human CD8	BD biosciences	Cat#612915; RRID:AB_2870200
BUV395, Mouse Anti-human CD56	BD biosciences	Cat#563555; RRID:AB_2687886
PE-Cy7, Anti-human TCR V-delta-1	Invitrogen	Cat#25-5679-42; RRID:AB_2762454
FITC, Anti-human TCR V-delta-2	Biolegend	Cat#331406; RRID:AB_1089230
BB700 Mouse Anti-human CD3	BD Biosciences	Cat#566575; RRID:AB_2860004
BV421, Anti-human TCR Va24-Ja18	Biolegend	Cat#342915; RRID:AB_2564004
Alexa Fluor® 700, Anti-human CD45	Biolegend	Cat#368514; RRID:AB_2566374
APC, Anti-human CD218a (IL-18Ra)	eBioscience™	Cat#17-7183-42; RRID:AB_2573258
FITC, Anti-human TCR V-delta-1	Invitrogen	Cat#TCR2730; RRID:AB_223624
BV421, Anti-human TCR a/b	Biolegend	Cat#306722; RRID:AB_2562805
PE, Rabbit Anti-human SIGIRR	Invitrogen	Cat#MA5-40972; RRID:AB_2898733
APC, Mouse Anti-Human IFN-γ	BD biosciences	Cat#554702; RRID:AB_398580
BV750, Rat Anti-human IL-2	BD Biosciences	Cat#566361; RRID:AB_2739710
PE-Cy7, Anti-Human CD107a (LAMP-1)	Biolegend	Cat#328618; RRID:AB_11147955
Human Interleukin-18 Binding Protein (IL-18BP)	Assaypro	Cat#30337-05161; RRID:AB_3720867
Ultra-LEAF™ Purified anti-human IL-18BP Antibody	Biolegend	Cat#947704; RRID:AB_2910511
Human IL-18/IL-1F4 Antibody	Biotechne	Cat#D044-3; AB_356964
Purified Rat IgG2a, λ Isotype Ctrl Antibody	Biolegend	Cat#402302; AB_3097078
Biological samples		
Patient derived organoids tumor/healthy (overview in Table S2).	The Netherlands Cancer Institute	NL48824.031.14
Primary human tumor/healthy tissue (colon/liver/CRC)	The Netherlands Cancer Institute	NL48824.031.14/IRBdm24-078
Healthy donor blood (buffy coat)	Sanquin blood bank	NA
Chemicals, peptides, and recombinant proteins		
B27 supplement without vitamin A	GIBCO	Cat#C12587-010
B27 supplement	GIBCO	Cat#17504-044
N-Acetylcysteine	Sigma-Aldrich	Cat#A9165-5G
Nicotinamide	Sigma-Aldrich	Cat#N0636
Human recombinant EGF	Peprotech	Cat#AF-100-15
A83-01	Tocris	Cat#2939
SB202190	Cayman Chemicals	Cat#10010399
Prostaglandin E2	Cayman Chemicals	Cat#14010-1
Y-27632	Sigma-Aldrich	Cat#Y-0503
Human recombinant FGF-7	Peprotech	Cat#100-19
Human recombinant FGF-10	Peprotech	Cat#100-26
Cultrex RGF BME, Type 2,	R&D/Biotechne	Cat#3536-005-02
Advanced DMEM-F12	GIBCO	Cat#12634-028
Penicillin/streptomycin	GIBCO	Cat#15070063
Ultraglutamine type I	Lonza	Cat#BE17-605E
HEPES	GIBCO	Cat#15630-056

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
TrypLE Express	GIBCO	Cat#12604-013
Human Serum	Sigma-Aldrich	Cat#H4522-100ML
IMDM, GlutaMAX TM Supplement	Thermofischer	Cat#31980048
AIM-V Medium (1X)	GIBCO	Cat#12055083
RPMI 1640	GIBCO	Cat#11875093
GolgiSTOP (Monensin)	BD biosciences	Cat#554724
GolgiPLUG (Brefeldin A)	BD biosciences	Cat#555029
Human BD Fc Block TM	BD biosciences	Cat#564219
Phorbol 12-myristate 13-acetate (PMA)	Sigma-Aldrich	Cat#19-144
Ionomycin	Sigma-Aldrich	Cat#I9657
Hanks' Balanced Salt solution HBSS	GIBCO	Cat#55021C-1000ML
Benzonase	Merck	Cat#70746-3
Pulmozyme® (dornase alfa)	Roche	Cat#13824856
Interleukin-2	Pharmacy Netherlands Cancer Institute	Proleukin
Human interleukin-12 p70	Peprotech	Cat#200-12-100UG
Human recombinant interleukin-15	Peprotech	Cat#200-15-100UG
Recombinant human IL-18 protein	Invivogen	Cat#rcyec-hil18
β-ethynylserine (βES)	Kindly provided by Bob Ignacio and Kim Bonger	
Halt TM Protease and Phosphatase Inhibitor Cocktail	Thermo Scientific	Cat#78440
Critical commercial assays		
LIVE/DEAD TM Fixable Near-IR Dead Cell Stain Kit, for 633 or 635 nm excitation	Invitrogen	Cat#L34976
LEGENDplex TM Human Inflammation Panel 1 (13-plex) with w/VbP	Biolegend	Cat#740809
Human IL-18BP ELISA Kit	Invitrogen	Cat#EHIL18BP
Incucyte® Caspase-3/7 Dye for Apoptosis	Sartorius	Cat#4440
Fixation/Permeabilization solution kit	Invitrogen	Cat#L10119
Anti-TCR γ/δ MicroBead Kit	Miltenyi	Cat#130-050-701
Deposited data		
Genomics, transcriptomics and clinical data metastatic cancer patients	Priestley et al. ²⁸	https://www.hartwigmedicalfoundation.nl/
Tissue immune cell atlas	Domínguez Conde et al. ²⁹	https://www.tissueimmunecellatlas.org/
Liver cell atlas	Guilliams et al. ^{30,45}	https://www.livercellatlas.org/
Proteomics secreted proteins PDTO's	This manuscript, Project accession PRIDE:PXD071538	https://www.ebi.ac.uk/pride/archive/projects/PXD071538 Perez-Riverol et al.
Software and algorithms		
Prism 10 Version 10.4.1 (532)	Graphpad Software	https://www.graphpad.com
FlowJo version 10.10.0	FlowJo	https://www.flowjo.com/
BD FACSDiva TM Software, version 8.0.2.	BD Biosciences	https://www.bdbiosciences.com/en-us/products/software/instrument-software/bd-facsdiva-software
iQue Forecyt Standard Edition 8.1 (8.1.7606)	Sartorius	–
LEGENDplex Data Analysis Software Suite	LEGENDPLEX	–
R (v4.2.2)	R core team	www.r-project.org

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Python (v3.8)	Python Software Foundation	www.python.org
Limma-voom (v.3.58.1)	Ritchie et al. ⁵⁵	https://bioconductor.org/packages/release/bioc/html/limma.html
DESeq2 (v.1.38.3)	Love et al. ⁵⁶	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
fgsea (v1.27.0)	Korotkevich et al. ⁵⁷	https://bioconductor.org/packages/release/bioc/html/fgsea.html
edgeR (v.4.0.16)	Robinson et al. ⁵⁸	https://bioconductor.org/packages/release/bioc/html/edgeR.html
PURPLE (v.3.7.1)	HMFtools suit, Priestley et al. ²⁸	https://github.com/hartwigmedical/pipeline5/releases
DIA-NN (version 1.9.2)	Demichev et al. ⁵⁹	https://github.com/vdemichev/DiaNN/releases
STAR (v2.7.10a)	–	https://github.com/alexdobin/STAR/releases
Subread package (v2.0.1)	Liao et al. ⁶⁰	https://github.com/ShiLab-Bioinformatics/subread

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Within the Hartwig Medical Foundation cohort, patients with advanced cancer without any further treatment options were included as part of the CPCT-02 (NCT01855477) and DRUP (NCT02925234) clinical studies, which were approved by the medical ethical committees (METC) of the University Medical Center Utrecht and the Netherlands Cancer Institute, respectively. Informed consent was given for WGS as well as data sharing for research purposes. Clinical analyses were performed using data from the DRUP study, which was approved by the Medical Ethical Committee of the Netherlands Cancer Institute in Amsterdam and is conducted in accordance with Good Clinical Practice guidelines and the Declaration of Helsinki's ethical principles for medical research. All patients were >18 years with a performance status 0–2, further baseline characteristics including gender and age, are provided in [Tables 1](#) and [S3](#). Written informed consent was obtained from all included patients. DRUP is registered with [ClinicalTrials.gov](https://clinicaltrials.gov), number NCT02925234.

For *ex vivo* tissue assays, tumor and nonmalignant tissue samples were collected from individuals with colorectal cancer undergoing surgical treatment at the Netherlands Cancer Institute-Antoni van Leeuwenhoekziekenhuis (NKI-AVL). The study was approved by the institutional review board of the NKI-AVL (IRBdm24-078) and performed in compliance with all relevant ethical regulations. All patients consented to research usage of material not required for diagnostic use either by opt-out procedure or via previous written informed consent (after 23 May 2018). Patient derived tumor organoids were previously generated, an overview of organoids is provided in [Table S2](#). Sampling was approved by the Medical Ethical Committee of the NKI-AVL (NL48824.031.14) and written informed consent was obtained from all of the patients. Usage of organoids for further research purposes was included in informed consent. Organoids were regularly checked for mycoplasma contamination and authenticated via SNP array as previously described.⁴⁹

METHOD DETAILS

Clinical data analysis

The Drug Rediscovery Protocol (DRUP) is an ongoing prospective, multicenter, nonrandomized clinical trial in which patients with advanced or metastatic solid tumors who have exhausted all standard-of-care options are treated based on their tumor molecular profile with approved targeted therapies or immunotherapies outside their registered indication.²⁶ For the clinical analyses, we included patients with MMR-d cancers treated with PD-(L)1 blockade from three completed cohorts with MMR-d/MSI-hi as an actionable target.^{14,61} For patients where WGS was available (which is all patients included in the transcriptomic-based biomarker analysis), microsatellite status was confirmed and patients with discordant results of of routine diagnostics were excluded from the clinical and biomarker analyses. Patients were considered evaluable according to study protocol if response was radiologically or clinically evaluable, if they received at least one (durvalumab) or two (nivolumab) treatment administrations. The presence of liver-metastases was determined based on (non)-target lesions according to RECIST v1.1 guidelines on baseline imaging. For the current analyses, the primary outcome measure was PFS, all assessed according to the RECIST v1.1 guidelines⁶² or Response Assessment in Neuro-Oncology (RANO).⁶³ For all LM patients with no clinical benefit, the disease progression pattern was analyzed. Upon initial RECIST 1.1 confirmed progressive disease, patients were classified as 'isolated liver progression' if patients 1) developed new hepatic lesions with no new lesions at other sites, or 2) progression of existing hepatic metastases but not at other sites. If patients displayed increases in tumor volume of extra-hepatic lesions, or developed new lesions outside of the liver, patients were

classified as 'systemic progression'. Tumor burden was assessed at baseline following RECIST v1.1 procedure, which takes the sum of the longest diameter for all 5 target lesions as an estimate.

For validation of clinical findings using real-world data, a datarequest was made to the Netherlands Cancer Registry, which was approved by its scientific committee. Response data were requested from all adult patients with MMR-d mCRC (diagnosed between 2018 and 2023) that received first-line ICB (Nivolumab, Pembrolizumab, Atetolizumab, Avelumab, Durvalumab) and where the presence of LM was determined at baseline. Patients were excluded from analysis if: 1) they underwent surgical treatment of liver metastases prior to immunotherapy, or 2) combination immunotherapy regimens were given (e.g., anti-PD1 + anti-VEGF or TKI + anti-PD1). Response to treatment was assessed by the treating physician. For these patients, mismatch repair status was determined via routine diagnostics, however, WGS data was not available for validation thereof.

Library preparation and NGS

Samples were processed as previously described in a standardized and centralized facility.²⁸ Briefly, DNA was extracted from biopsy and blood samples using an automated system (QiaSymphony) following the manufacturer's protocols (Qiagen). Blood DNA was isolated with the DSP DNA Midi kit, while tissue DNA was obtained using the QIAasymphony DSP DNA Mini kit. Prior to DNA extraction from tissue, biopsies were homogenized in 100 mL of nuclease-free water with the Qiagen TissueLyzer and divided into two equal portions for parallel automated DNA and RNA extraction using QiaSymphony. RNA was isolated with the QIAGEN QIAasymphony RNA kit, processed with the KAPA RNA Hyper + RiboErase HMR kit, and subjected to paired-end sequencing on either the Illumina NextSeq550 platform (2 × 75 bp) or the Illumina NovaSeq6000 platform (2 × 150 bp).

WGS analysis

Whole genome sequencing and subsequent data analysis was performed by Hartwig Medical Foundation as described previously,²⁸ whereby reads were mapped against the reference genome GRCh37. PURPLE (v.3.7.1), a tool from the HMFtools suit, was used to calculate tumor mutational load (# of missense variants per sample) and tumor mutational burden (# of passing variants per Mb).

RNA data analysis

RNA-seq data was processed using Cutadapt to retain reads longer than 35 nucleotides and to remove TruSeq adapters. The filtered reads were then aligned to the GRCh38 reference genome (Gencode v35) using STAR (v2.7.10a) in 2-pass mode and quantified with featureCounts from the Subread package (v2.0.1)⁵⁰.

The R packages edgeR⁵⁸ (v.4.0.16) and limma⁵⁵ (v.3.58.1) were used to test differentially abundant genes. First, *normLibSizes* function from EdgeR was used to calculate the normalization factors to generate log2 counts per million reads (CPM) with the function *voom* from limma. Next, differential expression of genes was calculated using a linear Limma model with empirical Bayes smoothing of standard errors. A contrast matrix was generated to define pertinent comparisons. The following design formula was used for the linear model: *Expression* ~ *CB_LM* + *lymphnode* + [*intercept*]. Here *CB_LM* was a factor that combined LM and clinical benefit (CB) status. Due to the large amount of different tumor types in the cohort, tumor type was not included as covariate in the linear model. However, we corrected for lymph node biopsy sampling, by the categorical factor *lymphnode*, as this variable could obscure the results related to immune cell genes and signatures.

The same approach was followed for differential abundance analysis between IL18-high (higher 1/3 expressing samples) and IL18-low patients (bottom 1/3 expressing samples). In this case the following design formula was used: *Expression* ~ *IL18.status* + *lymphnode* + *purity* + *tumorGroups*. Genes were ranked based on log fold change difference between IL18-high and IL18-low samples to perform gene set enrichment analysis. This was done with the *fgsea()* function from the *fgsea* package⁵⁷ (v.1.28.0), and gene sets from the Hallmark collection retrieved with *msigdbR* (v.7.5.1) package.

Single cell analysis

Publicly available data was used for the single cells analysis. Processed and annotated immune cell data from the Human cell atlas data portal²⁹ was downloaded to visualize gene expression for GD1 cells from the gut (Gut-VD1) from liver (Liv-VD1) or other organs (Non-Gut-VD1). We first selected cells manually annotated as Tissue-Resident Memory gd-cells (*Trm_Tgd*) or CRTAM-expressing gd-cells (*Tgd-CRTAM+*) from the dataset. We next selected cells expressing *TRDV1*, a gene encoding for the VD1+ GD TCR. We then classified the gd-cells based on the metadata variable *Organ*. Cells originated from the caecum (CAE), duodenum (DUO), ileum (ILE), proximal jejunum (JEJUP), lower proximal jejunum (JEJLP), sigmoid colon (SCL), and transverse colon (TCL) were classified as Gut-VD1, liver (LIV) were classified as Liv-VD1, whereas bone marrow (BMA), and lung (LNG) were classified as other-VD1.

Tissue sample processing

Solid tumor lesions were macroscopically selected from the resected tumor material and parts of the tissue were collected in cold collection medium Ad-DF+++ (Advanced DMEM/F12 (GIBCO) supplemented with 2 mM Ultraglutamine I (Lonza), 10 mM HEPES (GIBCO), and 100/100 U/ml Pencillin/Streptomycin (GIBCO)). Tissue materials collected were immediately processed by manual mechanic dissociation into small fragments. After processing fragments were frozen in cryovials containing 1 mL of FBS with 10% dimethyl sulfoxide (Sigma-Aldrich). All tissue fragments were cryopreserved in liquid nitrogen until further usage.

Digestion of tissue samples and flowcytometric analysis of tissue-resident immune cells

Vials with cryopreserved tissue fragments were thawed in a water bath at 37°C, and subsequently washed in HBSS (Gibco) warm medium by flushing them multiple times on a cell strainer in a 6-well plate. Next, the fragments were processed into single-cell suspensions by enzymatic digestion in digestion mix (HBSS (Gibco) with 1% penicillin-streptomycin (Gibco), 12.6 $\mu\text{g mL}^{-1}$ Pulmozyme (Roche) and 1 mg mL^{-1} collagenase type IV (Sigma-Aldrich)) for 45–60 min at 37°C and under slow rotation. Enzyme activity was neutralized by addition of ice-cold HBSS. Samples were then washed with HBSS (Gibco), filtered over a 40- μm filter mesh, and remaining tissue structures were mashed through. Epithelial cell fraction was next removed by centrifugation (50g for 4 min) and cell debris was removed via density gradient separation with the debris removal kit (Miltenyi biotech). For immunophenotyping, cells were directly resuspended in 50 μL of PBS and incubated with Fc receptor blocking agent (eBioscience), followed by viability staining with Near IR viability dye (Life technologies) for 20 min at 4°C. Cells were then washed, resuspended in 150 μL of staining buffer (PBS, 0.5% bovine serum albumin (Sigma-Aldrich), EDTA) containing the antibodies and incubated for 20 min at 4°C. The following antibodies were used: CD4-BUV805 (BD biosciences, 1:200), CD8-BUV563 (BD Biosciences, 1:200), CD56-BUV395 (BD Biosciences, 1:100), VD1-TCR-PE-Cy7 (Invitrogen; 1:40), VD2-TCR-FITC (Biolegend 1:50), CD3-BB700 (BD biosciences, 1:160), TCR VA24-Ja18-BV421 (Biolegend, 1:50), CD45-AF700 (Biolegend, 1:50), IL18R1-APC (eBiosciences, 1:50).

For cytokine stimulation experiments, cells were cultured for 24 h in $\gamma\delta$ T cell medium (IMDM (Gibco) + 10% human serum (Sigma-Aldrich), supplemented with 150 u/ml IL-2 and 5 ng/mL IL-15) with the addition of cytokine combinations of interest, namely recombinant IL-12 (Peprotech, 10ng/ml) and IL18 (Peprotech, 10ng/ml). Part of the supernatant was harvested for cytokine assays. Next, Golgi-Plug (1:1000, BD) and Golgi-Stop (1:1500, BD) was added culture continued for an additional 3 h. Next, cells were washed twice and resuspended in 50 μL of PBS and incubated with Fc receptor blocking agent (eBioscience) and subsequently LIVE/DEAD IR Dye (Life Sciences) for 20 min at 4°C each. The following antibodies were used for staining: VD1-TCR-FITC (Invitrogen, 1:20), TCR-AB-BV421 (Biolegend, 1:40), CD45-AF700 (Biolegend, 1:50), SIGIRR-PE (Invitrogen, 1:50), IFN γ -APC (BD biosciences, 1:40), IL-2-BV750 (BD biosciences, 1:50), CD107-PE-Cy7 (Biolegend, 1:50). Prior to staining, cells were then washed, resuspended in 50 μL of staining buffer (PBS, 0.5% bovine serum albumin (Sigma-Aldrich), EDTA) containing the aforementioned antibodies and incubated for 20 min at 4°C. Cells were washed twice, followed by fixation and permeabilization using the Cytofix/Cytoperm kit (BD Biosciences), then staining of intracellular proteins. All experiments were performed using a BD FACSymphony cytometer using Diva software. Data analyses used FlowJo Software version 10.10.

Analysis of inflammatory cytokines in patient plasma

Patient plasma samples were collected as a part of the DRUP-study²⁶ and analyzed via the Legendplex Human Inflammation Panel 1 (BioLegend) according to manufacturer's instructions. Plasma samples were acquired pre-treatment from whole blood samples (Streck). Readouts were performed using a iQue 3 (Sartorius) with standard Forecyt software. Analyses of flow cytometry data was performed in the provided LEGENDplex Data Analysis Software Suite.

For measuring IL18BP in patient plasma, the same samples were used as for IL18 measurements. Measurements were made using Human IL-18BP ELISA Kit (ThermoFisher scientific #EHIL18BP) according to manufacturer's instructions. Tecan Infinite M200 plate reader was used for readout of the absorbance at 450 nm.

Organoid culturing

Tumor organoids were derived from MMR-d CRC patients from resection specimens as previously reported.^{48,49} Patients were eligible if routine molecular testing demonstrated MMR-d by loss of staining of the mismatch repair proteins (MLH1, MSH2, MSH6 and PMS2) by immunohistochemistry (IHC) or MSI-H by either polymerase chain reaction (PCR), panel-based next generation sequencing (NGS) or whole genome sequencing (WGS) of the original patient tumor. Sampling was approved by the Medical Ethical Committee of the NKI-AvL (NL48824.031.14) and written informed consent was obtained from all of the patients. In brief, tumor tissue was mechanically dissociated and digested with 1.5 mg mL^{-1} of collagenase II (Sigma-Aldrich), 10 $\mu\text{g mL}^{-1}$ of hyaluronidase type IV (Sigma-Aldrich) and 10 μM Y-27632 (Sigma-Aldrich). Cells plated in Cultrex RGF BME type 2 (3533-005-02, R&D systems) and placed into a 37°C incubator for 20 min, and CRC organoid medium was added. The medium was used as previously described, and is composed of Ad-DF+++ (Advanced DMEM/F12 (GIBCO) supplemented with 2 mM Ultraglutamine I (Lonza), 10 mM HEPES (GIBCO), 100 U mL^{-1} of each penicillin and streptomycin (GIBCO), 10% noggin-conditioned medium, 20% R-spondin1-conditioned medium, 1 $\times$ B27 supplement without vitamin A (GIBCO), 1.25 mM N-acetylcysteine (Sigma-Aldrich), 10 mM nicotinamide (Sigma-Aldrich), 50 ng mL^{-1} human recombinant EGF (Peprotech), 500 nM A83-01 (Tocris), 3 μM SB202190 (Cayman Chemicals) and 10 nM prostaglandin E2 (Cayman Chemicals). Organoids were passaged every week, by enzymatical dissociation with TrypLE Express (Gibco), washing and subsequent embedding in BME. Organoids were authenticated by SNP array, were tested on a monthly basis for Mycoplasma using Mycoplasma PCR43 and the MycoAlert Mycoplasma Detection Kit (LT07-318). In the first two weeks of organoid culture, 1 $\times$ Primocin (Invivogen) was added to prevent microbial contamination.

Sorting and expanding of $\gamma\delta$ T cells from healthy donor PBMC

Peripheral blood mononuclear cells (PBMCs) were isolated from leukopacks from healthy donors using Ficoll-Paque density gradient separation (Lympholyte-H Cell Separation Media, Tebubio) and cryopreserved until later use. PBMCs were thawed in pre-warmed T cell thawing medium composed of RPMI 1640 (Gibco), supplemented with 2 mM Ultraglutamine I (Lonza),

100/100 U/ml Penicillin/Streptomycin (Gibco), and 10% FC. Cells were subsequently treated with 25 U/ml benzonase (Merck) for 15 min, washed and resuspended in 2×10^6 cells/ml T cell culture medium (IMDM 1640 supplemented with 2mM Ultraglutamine I, 100/100 U/ml Penicillin/Streptomycin, and 10% human serum) in presence of 150 U/ml IL-2 (Peprotech). Next, $\gamma\delta$ T cells were pre-enriched prior to sorting using TCR γ/δ T cell Isolation Kit (Miltenyi biotec) using magnetic assisted cell sorting, according to manufacturers protocol. Next, the whole gamma delta T cell pool was stained with NiR live dead, CD3-BB700 (BD Biosciences, 1:160), TCR-AB-BV421 (Biolegend, 1:40), VD1-TCR-PE-Cy7 (Invitrogen; 1:40) for 25 min at 4°C. Single, live, CD3⁺ TCR α/β -, TCR YD+, V δ 1+ cells were sorted on the FACS Aria III 4L (BD Biosciences) under sterile conditions for further cell culturing. Cells were subsequently expanded in medium containing irradiated feeder cells. 1 μ g phytohemagglutinin-L (PHA), 1000 U/ml IL-2 and 10 ng/mL IL-15 for two to three weeks. Medium with cytokines was refreshed after initial 7 days, and subsequently every 2 days. After expansion protocol, purity of the V δ 1⁺ $\gamma\delta$ T cells was assessed via flow cytometry and cells were cryopreserved in FCS with 10% DMSO.

Killing assay

To assess gamma-delta T cell-mediated killing of tumor organoids and cell lines, organoids and cell lines were previously transduced with luciferase reporter gene using pLenti CMV Puro LUC (w168-1; Plasmid #17477; Addgene). Three days prior to killing assay, PBMC expanded $\gamma\delta$ T cells were thawed and rested for three days in $\gamma\delta$ T cell medium (IMDM +10% human serum, supplemented with 150 u/ml IL-2 and 5 ng/mL IL-15). After three days, organoids were isolated from geltrex and dissociated to single cell and counted. Cells were seeded at a 1:1 effector to target ratio in $\gamma\delta$ T cell m cell medium. For organoids, 50.000 cells each were plated per well, for the cell line, 25.000 cells each were plated per well. To assess the effect of different cytokines on killing-capacity, recombinant IL-12 (10ng/ml) and IL18 (10ng/ml) was added (remaining present throughout the remaining co-culture). After 24 h, tumor cell viability was assessed by luciferase reporter assay using 3 μ g/mL luciferin and measured with a Tecan reader (1000 ms exposure).

For incucyte based immune cell killing assays, Organoids were dissociated with TrypLE Express Enzyme (ThermoFisher), and 50,000 cells were plated in $\gamma\delta$ T cell medium of IMDM supplemented with 1:1000 Pen Strep, (ThermoFisher scientific), 1:1000 Stable Glutamine (Capricorn scientific) and 10% human serum. Cells were pre-stimulated with 20 ng/mL IFN γ and incubated with 1 μ g/mL Ultra-LEAF Purified anti-human IL18BP Antibody (BioLegend) or 1 μ g/mL Ultra-LEAF Purified Human IgG4 Isotype Ctrl Recombinant Antibody (BioLegend) for 24 h. After 24 h, 20,000 cells in GDT cell medium were plated. The cells were stimulated with 10 ng/mL IL-12 (PeproTech) and 1 ng/mL IL18 (Peprotech). To visualize killing, CellEvent Caspase-3/7 Green ReadyProbes Reagent (ThermoFisher scientific) was added in a concentration of 30 μ L per 1 mL medium. Immediately after this, plates were placed in the Sartorius Incucyte S3 Live-Cell Analysis System. Sartorius Incucyte software was used with standard scan type protocol settings. One scan per well was made every hour for 24 h. Background signal measured in wells where only IFN- γ was added to the tumor organoids was subtracted from the conditions. Cancer cells alone and cancer cells alone with caspase-3/7 were used as negative controls.

Reactivity assay for gamma delta T cell – organoid co-culture

For the cytokine stimulations during mono- and co-culture, PBMC expanded $\gamma\delta$ T cells were cultured in presence or absence of PDTO MSI-3 in $\gamma\delta$ T cell medium (IMDM +10% human serum), supplemented with IL-2 (150u/ml). Cells were stimulated with IL18 (1 ng/mL), IL12 (10ng/ml) or both for 24H. For the IL18BP-blocking experiment, cells were pre-stimulated with low-dose IFN γ (20ng/ml) prior to co-culture, and subsequently cultured for 24 h in IMDM supplemented with 10% human serum, IL-2 (150 u/ml), IL-12 (10ng/ml), and IL18BP Blocking antibody (Biolegend, 1 μ g/mL) or isotype control (Biolegend, 1 μ g/mL). After 24 h, 1 μ g/mL recombinant IL18 was added to each well and incubated for 1 h. Next, gamma delta T cells from different donors were added to the different conditions (at 50.000 cells per well). For the IL18 blocking experiment, PBMC expanded $\gamma\delta$ T cells were co-cultured together with PDTO MSI-3 in $\gamma\delta$ T cell medium (IMDM +10% human serum), supplemented with IL-2 (150u/ml) and IL12 (10ng/ml), in the presence or absence of an IL18 blocking antibody (MBL, clone 125-2H, 0.1 μ g/ml).

For all experiments, cells were stimulated with 50 ng/mL of phorbol-12-myristate-13-acetate (Sigma-Aldrich) and 1 μ g/mL of ionomycin (Sigma-Aldrich) as a positive control. After 1 h of incubation at 37°C, GolgiSTOP (BD Biosciences, 1:1500) and GolgiPlug (BD Biosciences, 1:1,000) were added. After 4 h of incubation, cells were washed and incubated with human Fc receptor block (BioLegend, 1:200) for 20 min at 4°C. Next, cells were stained with cell surface antibodies: CD3-BB700 (BD Biosciences, 1:160), V δ 1 TCR – FITC (Invitrogen, 1:20), AB TCR-BV421 (Biolegend, 1:40), and a 1:1,000 near-infrared viability dye (Life Technologies). Next, cells were fixed and permeabilized using fixation buffer and intracellular staining permeabilization wash buffer (BioLegend), and cells were subsequently stained with an anti-IFNG-APC (Biolegend, 1:40) for 20 min at 4°C. Compensation was performed using CompBeads (BD Biosciences) and flow cytometry was performed on a BD FACSymphony cytometer using Diva software. Data analyses used FlowJo Software version 10.10.

Sample preparation for secretomics

Secretomics was performed using a previously published metabolic labeling method based on threonine-derived non-canonical amino acid tag (THRONCAT).⁶⁴ In brief, four CRC MMR-d/MSI organoids were plated in complete CRC culture media in duplicate at 1.6×10^6 cells per replicate. Five days later, 0.5 mM β -ethynylserine (β ES) was added to complete culture media for 24 h to metabolically label newly synthesized proteins. Simultaneously, 20 ng/mL IFN γ was added to half of the samples. After 24 h, culture media was collected and centrifuged for 5 min at 1,000 $\times$ g to pellet remaining cells. Supernatant was collected and protease inhibitor (1:100,

Halt™), and phosphatase inhibitor (1:100, Halt™) were added to the culture media before storing the samples at -80°C . Input for secretome analysis was normalized to total protein amount, measuring the optical density of the total protein content of each sample on a Beckman Coulter machine. For secretome analysis, culture media were processed and measured as previously described,⁶⁵ with the exception that, depending on the volume of medium collected, Amicon Ultra 3kDa cutoff 15mL or 50mL ultrafiltration spin filters (Merck Millipore UFC900324; UFC800308) were used according to the manufacturer's protocol.

Mass spectrometry

Prior to mass spectrometry analysis, the peptides were reconstituted in 2% formic acid. Peptide mixtures were analyzed on the Orbitrap Astral mass spectrometer (Thermo Scientific) connected to a Vanquish Neo nano-LC system (Thermo Scientific). The Vanquish Neo was operated in the trap-and-elute mode and peptides were loaded onto a Pepmap 100 C18 5 μm trap column (300 μm $\times$ 5mm, Thermo Scientific), before separation on the analytical column (AUR325075C18TS, 1.7 μm /75 μm $\times$ 250mm, Ionopticks or ES906, 2.0 μm /150 μm $\times$ 15cm, Thermo Scientific). Solvent A was 0.1% formic acid/water and solvent B was 0.1% formic acid/80% acetonitrile. Peptides were analyzed in a 30 samples-per day (30SPD) LC-MS method with a non-linear gradient. The Orbitrap Astral was run in data-independent acquisition (DIA) mode.

Secretomics data analysis

Raw data were analyzed by DIA-NN (version 1.9.2)⁵⁹ without a spectral library and with "Deep learning" option enabled. The Swissprot Human database (DIA-NN 1.9.2: 20,435, release 2024_08) was added for the library-free search. The Quantification strategy was set to Robust LC (high accuracy) for DIA-NN version 1.8.1 and for DIA-NN version 1.9.2 quantification strategy was set to QuantUMS (high precision). MBR option was enabled and all other settings were kept at the default values. The protein groups report from DIA-NN was used for downstream analysis in Perseus (version: 2.0.10.0 or later).⁶⁶ Values were Log2-transformed, after which proteins were filtered for at least 66% valid values in at least one sample group. Missing values were replaced by imputation based a normal distribution using a width of 0.3 and a minimal downshift of 2.4. To focus on truly secreted proteins, rather than aspecific release due to cell death, we filtered out proteins not annotated as secreted according to the human protein atlas.⁶⁷ Differentially expressed proteins were determined using a mixed linear model with correction for replicates (minimal threshold: adjusted p value ≤ 0.05). Differential expression of IL18BP between CRC MSI organoids $\pm$ stimulation with IFN γ was calculated using a paired Student's t test.

IL18BP ELISA PDTO

In brief, three CRC MMR-d/MSI and one healthy colon organoids were plated in complete AdDF+++ in duplicate at 1.0×10^6 cells per replicate, with 20 ng/mL IFN γ for 24 h. Next day, cells were moved via centrifugation and filtration, and conditioned medium was harvested and cryopreserved until assay. IL18BP measurements were performed using ELISA was performed using Human IL-18BP ELISA Kit (ThermoFisher scientific #EHIL18BP) according to manufacturer's instructions. Tecan Infinite M200 plate reader was used for readout of the absorbance at 450 nm.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses for clinical and bioinformatics analyses were performed using R version 4.2.2 or Python version 3.8. Patient characteristics, and tumor responses were summarized using descriptive statistics. Differences in CB between metastatic groups were calculated using the Fisher exact test (categorical variables), Wilcoxon test (continuous variables), and linear-by-linear association test (ordinal variables). Kaplan–Meier methods were used to estimate PFS (calculated from the start of treatment to disease progression or death from any cause and censoring patients alive without progression). For differential gene expression analyses, two-sided limma-voom-based regression were used with multiple testing correction.

For translational and *in vitro* analyses, data were analyzed using GraphPad Prism version 10.4.1. Group sizes and definition of error bars is indicated in figure legends. Statistical analysis was performed using a two-tailed Student's t test. p values <0.05 were considered significant; significance values are indicated as * ($p < 0.05$), ** ($p < 0.01$) and *** ($p < 0.001$).